ENGINEERING ECONOMICS

Principles of Engg Economy

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Engineering Economy

- Engineering economy involves in **collection of techniques** that simplify comparisons of alternatives on an economic basis .

- Engineering economics begins only after the alternatives have been identified.

PRINCIPLES OF ENGINEERING ECONOMY

Once a problem/need has been clearly defined, the foundation of the discipline (Engineering Economics) can be discussed in terms of following seven principles.

1. Develop the Alternatives
2. Focus on the Differences
3. Use a Consistent Viewpoint
4. Use a Common Unit of Measure
5. Consider All Relevant Criteria
6. Make Uncertainty Explicit
7. Revisit Your Decisions

Example-P of Engg ECO.ppt

DEVELOP THE ALTERNATIVES

The choice (decision) is among alternatives. The alternatives need to be identified and then defined for subsequent analysis.

- Carefully define the problem.

- A decision situation involves making a choice among two or more alternatives.

- Developing and defining the alternatives for detailed evaluation is important because of the resulting impact on the quality of the decision.

- Engineers and managers should place a high priority on this responsibility.

- Creativity and innovation are essential to the process.

- One alternative that may be feasible in a decision situation is making no change to the current operation or set of conditions (i-e doing nothing). If you consider this option feasible, make sure it is considered in the analysis.

Example-P of Engg ECO.ppt

Example: We need to buy a Car.

Alternatives are..

1. Corolla

2. Civic

3. Prius

FOCUS ON THE DIFFERENCES

Only the differences in expected future outcomes among the alternatives are relevant to their comparison and should be considered in the decision.

- The differences in the future outcomes of the alternatives are important to consider.

- If all future outcomes of the feasible alternatives were exactly the same, there would be no basis or need for comparison. We would be indifferent among the alternatives and could make a decision using a random selection.

- Outcome that are common to all alternatives can be disregarded in the comparison and decision. Example-P of Engg ECO.ppt

For Example :

Specifications	Corolla	Civic	Prius
Price	PKR: 3.2 Million	PKR: 3.2 Million	PKR: 3.2 Million
Engine Capacity	1800 cc	1800 cc	1800 cc
Transmission	Automatic	Automatic	Automatic
Sitting Capacity	5 P	5 P	5 P
Fuel Average	13 Km/L	11 Km/L	19 Km/L
Safety	2 Airbags	2 Airbags	7 Airbags
Performance	172 Hp	179 Hp	152 Hp
Resale value after 2 y	PKR: 2.2 M	PKR: 1.9 M	PKR: 2.5 M

USE A CONSISTENT VIEWPOINT

The prospective outcomes of the alternatives, economic and other, should be consistently developed from a defined viewpoint (perspective).

- The perspective of the decision maker, which is often that of the owners of the firm, would normally be used. However, it is important that the viewpoint for the particular decision be first defined and then used consistently in the description, analysis and comparison of the alternatives.

- In certain situations, the viewpoint may not be that of the owners of the firm, it may be of employees. Example-P of Engg ECO.ppt

For Example :



Specifications	Corolla	Civic	Prius
Price	~	~	~
Engine Capacity	~	~	~
Transmission	~	~	~
Sitting Capacity	~	~	~
Fuel Average	~	~	~
Safety	~	~	~
Performance	~	~	~
Resale value after 2 y	~	~	~

USE A COMMON UNIT OF MEASURE

Using a common unit of measurement to enumerate as many of the prospective outcomes as possible will make easier the analysis and comparison of alternatives.

- It is desirable to make as many prospective outcomes as possible; directly comparable.

- For economic consequences, a monetary unit such as dollars is the common measure.

- Try to translate other outcomes (which do not initially appear to be economic) into the monetary unit.

- The outcomes that are not economic and can not be converted to monetary units, must be described explicitly, so that the information is useful to the decision maker in the comparison of alternatives.

Example-P of Engg ECO.ppt

For Example :

Specifications	Corolla	Civic	Prius
Price	PKR	PKR	PKR
Engine Capacity	cc	cc	cc
Transmission	Automatic	Automatic	Automatic
Sitting Capacity	Persons	Persons	Persons
Fuel Average	Km/L	Km/L	Km/L
Safety	No. of Airbags	No. of Airbags	No. of Airbags
Performance	Horsepower	Horsepower	Horsepower
Resale value after 2 y	PKR	PKR	PKR

CONSIDER ALL RELEVANT CRITERIA

Selection of a preferred alternative (decision making) requires the use of a criterion (or several criteria). The decision process should consider the outcomes enumerated in the monetary unit and those expressed in some other unit of measurement or made explicit in a descriptive manner.

- The decision maker will normally select the alternative that will best serve the long term interests of the owner(s) of the organization.

- In engineering economic analysis, the primary criterion relates to the long term financial interests of the owners. This is based on the assumption that available capital will be allocated to provide maximum monetary return to the owners.

- There are other organizational objectives, that one like to achieve with the decision, and these should be considered and given weight in the selection of an alternative. These become the additional criteria in the decision making process.

Example-P of Engg ECO.ppt

For Example :



Specifications	Corolla	Civic	Prius
Price	PKR	PKR	PKR
Engine Capacity	cc	cc	cc
Transmission	Automatic	Automatic	Automatic
Sitting Capacity	Persons	Persons	Persons
Fuel Average	Km/L	Km/L	Km/L
Safety	No. of Airbags	No. of Airbags	No. of Airbags
Performance	Horsepower	Horsepower	Horsepower
Resale value after 2 y	PKR	PKR	PKR
Country of Assembly	Local	Local	Local
Registration PKr	80 K	85 K	205 K
Token Tax PKr	15 K	16 K	25 K

MAKE UNCERTAINTY EXPLICIT

Uncertainty is inherent in projecting (or estimating) the future outcomes of the alternatives and should be recognized in their analysis and comparison.

Explicit:

Precisely and clearly expressed or readily observable; leaving nothing to implication

- The analysis of the alternatives involves estimating the future consequences associated with each of them. The magnitude and the impact of future outcomes of any course of action are uncertain.

- Even if the alternative involves no change from current operations, the probability is high that today's estimates will not be what eventually occurs.

Example-P of Engg ECO.ppt

For Example :

Specifications	Corolla	Civic	Prius
Spare Parts availability	Easy	Easy	Difficult
Spare Parts Price	Cheap	Expensive	Expensive
Maintenance Cost	Cheap	Cheap	Expensive
Built Quality	Good	Relatively Low	Excellent
Resale value after 2 y	PKR: 2.2 M	PKR: 1.9 M	PKR: 2.5 M
Booking Time	3 Months	Readily Available	2 Months Import period
Registration PKr	80 K	85 K	205 K
Token Tax PKr	15 K	16 K	25 K

REVISIT YOUR DECISIONS

Improved decision making results from an adaptive process; to the extent practicable, the initial projected outcomes of the selected alternative should be subsequently compared with actual results achieved.

- Learning from and adapting based on our experience are essential for good organization.

- The evaluation of results versus the initial estimate of outcomes for the selected alternative is often considered impracticable or not worth the effort. Too often, no feedback to the decision making process occurs.

- Organizational discipline is needed to ensure implemented decisions are routinely post-evaluated and the results used to improve future analysis of alternatives and quality of decision making.

Example-P of Engg ECO.ppt

For Example :

Specifications	Corolla	Civic	Prius
Best Option			
Review & Revisit a	Now Best Option		Imports policies
Review & Revisit b	New Model Coming Soon	Now Best Option	

ENGINEERING ECONOMIC ANALYSIS PROCEDURE

1. Problem recognition, formulation and evaluation.
2. Development of the feasible alternatives.
3. Development of the cash flows for each alternative.
4. Selection of a criterion (or criteria).
5. Analysis and comparison of the alternatives.
6. Selection of the preferred alternative.
7. Performance monitoring and post-evaluation results.

Discussion